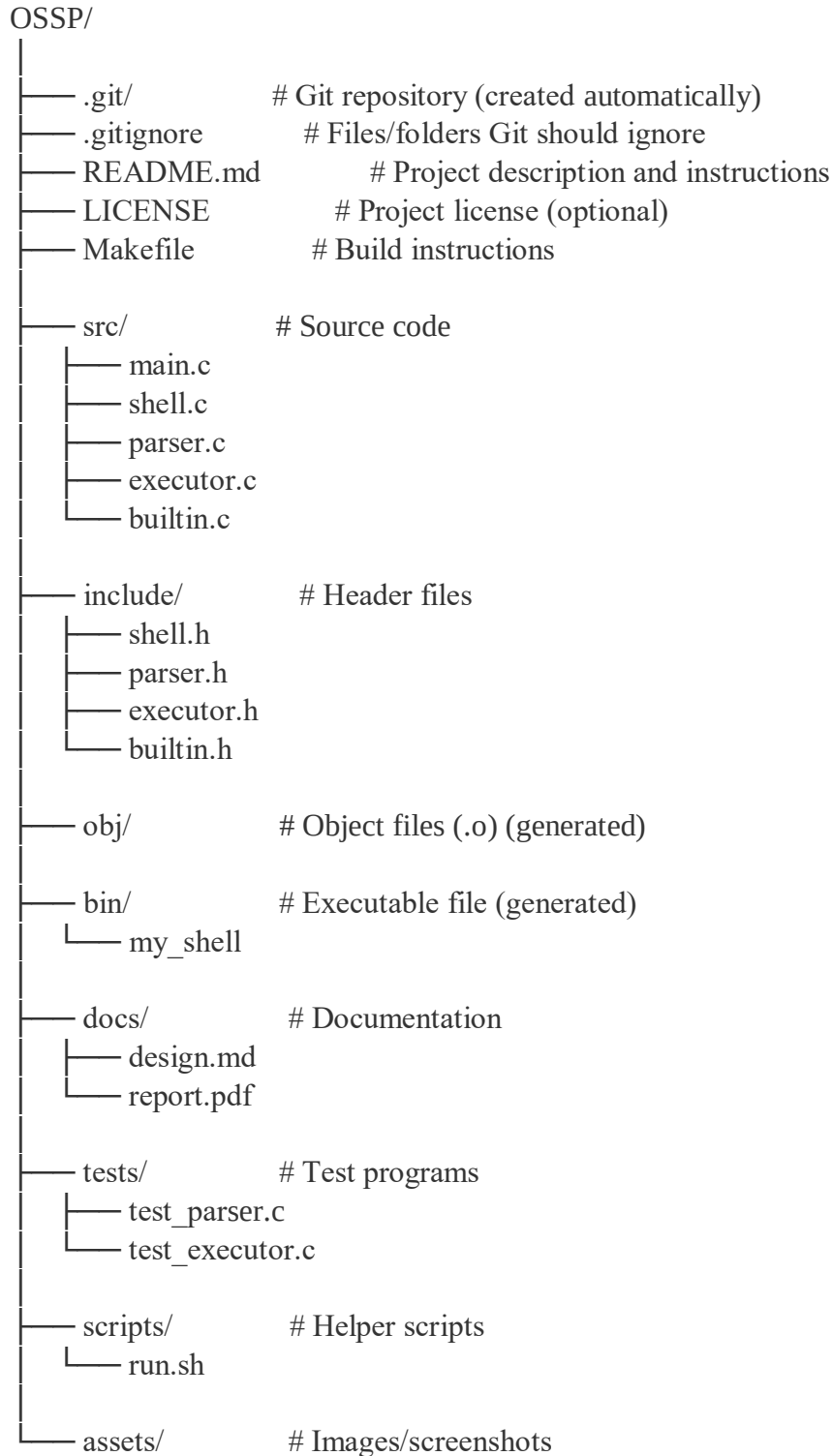


1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.

Task -1: Create Project Structure as follows:



└── architecture.png

Note: For creating this structure use mkdir and touch commands.

2. To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing.

Task-1: Using a manual command understand the fork() and exec() system calls

Task-2: Write a C program to create a new process and accepting the user commands using fork() and exec() system calls.

Example:

```
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;
    pid = fork(); // Create a new process
    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        // Child Process
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else {
        // Parent Process
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }
    return 0;
}
```